

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE423: Computer Graphics

Assignment 01 | Total Marks: 20 | Spring 2025

Q1. For a line segment from (5,-15) to (0,-5):

- a) Using the DDA (Digital Differential Analyzer) line drawing algorithm, compute all the pixels. Show all the steps at each stage. [4]
- b) Using the Midpoint line drawing algorithm, compute all the final pixels of the original zone. Show the present value of **d** and **Δs** (derivatives, updating of **d**) at each stage. Apply 8 way symmetry (Zone conversion) if required. [4]

Q2. Answer the following questions:

- a) What is the significance of the AND operation in the Cohen-Sutherland Algorithm? [2]
- b) Find the coordinates (x,y) of the point $t = 0.3$ for the line (1,3) to (7,10) using the parametric line equation. [2]
- c) A viewing window from (-50,-10) to (60,100) is given. Check whether the line segment (-50,-70) to (40,100) is trivially accepted or trivially rejected or partially inside using the
 - i) Cohen-Sutherland Algorithm. [4]
 - ii) Cyrus-Beck Algorithm (Calculate the value of t). [4]